

ADVANCED DETECTION WALKING STICK

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ABSTRACT

In this project we developed a collapsible walking stick that is designed to assist individuals with vision impairments by detecting objects in its user's path and notifying its user with a varying degree of vibrations. The stick will use ultra sonic sensor to detect objects based on their location; the stick will vibrate based on how close the object is. The project uses a raspberry pi zero to run the python code for the HC-SR04 ultrasonic sensor and coin vibration motors. A parallel breadboard circuit with a transistor and a resistor ensured the motors received stable power without overloading the Raspberry Pi. Future improvements to this project would be reducing the startup time and increasing the field of view of what the sensors detect.

INTRODUCTION

In the world, navigating is a challenge for individuals with visual impairments. Our goal is to make it easier for them by creating an advanced detection walking stick. The stick will be able to detect objects without touching them, which minimizes accidents. The way the stick works is it uses Ultrasonic sensors to detect the objects in front of the use. Then, based on how close it is to the sensor, its motors will vibrate and as it gets closer, vibrate with more intensity. This will run in real time for intuitive feedback. The raspberry pi will control everything, and it is powered by a small power bank that will be attached to the stick. The stick will be designed so it can run everything on its own, which means it doesn't require the internet or external devices. The design would be lightweight, portable, and easy to use, which will give the user the confidence they need in navigating environments safely.

SYSTEM SETUP

Problem Statement: Design a walking cane that detects objects in the user's path and provides a warning through vibration.

Requirements

- Ultrasonic sensor and buzzer/rumble motor function when switch is in the ON position
- Sensors should be able to detect objects within 1 second.
- The sensor and buzzer should function wirelessly.
- As the sensors get closer to the object, at 1ft it starts to rumble at 1 rumble a second. Within 1ft the speed of the rumble increases by 50% every inch.

Functional Decomposition

Level 0:	
Module	Collapsible Stick
Functionality	The walking stick can be collapsed to less than 60% of its original extended length.
Level 1:	
Module	Detection System
Inputs	Detection: Object Ultrasonic Sensor: detects object within 1ft
Outputs	Rumble Motor: Emits vibration after Sensors detects object
Functionality	Sensor and motor detect objects and emit vibration

DESIGN

Software Design

The code is written in Python, and it is run on a Raspberry Pi 0, which will communicate with the two HC-SR04 Ultrasonic Sensors. The code checks if the object is in the range of one of the sensors. If the object is in both detection areas of both sensors, then it will take which sensor is the closest. When the object is seen, the coin motors will vibrate, and the intensity is based on how close the object is.

Hardware Design

When designing this stick, we wanted it to be collapsible. The goal was to have the stick collapse to about 50% of its original length. In addition to this, we also knew we had to mount the sensors, motors, Raspberry pi 0, and breadboard onto the stick as well. We mounted these by using mounting tape and zip ties which, when using these two together, ensured our components were securely mounted onto the walking stick. For the breadboard we decided to use a parallel circuit, this way all the motors will be supplied with the same amount of voltage. This caused them to vibrate with the same intensity. On the breadboard we used a transistor and a 10-kilo ohm resistor to make the circuit run smoothly and we did not have to worry about overloading the Raspberry Pi 0.

RESULTS

The stick was able to detect objects in front of the user and was able to increase intensity as the object get closer. Unfortunately, we could not 3D print the stick due to the printers becoming unavailable, so we had to buy a walking stick on Amazon. This made the stick less durable, and we were unable to place the battery on it. The total budget of the project in end was \$90.



Figure 1 - Fully completed stick



Figure 2 - How components were mounted

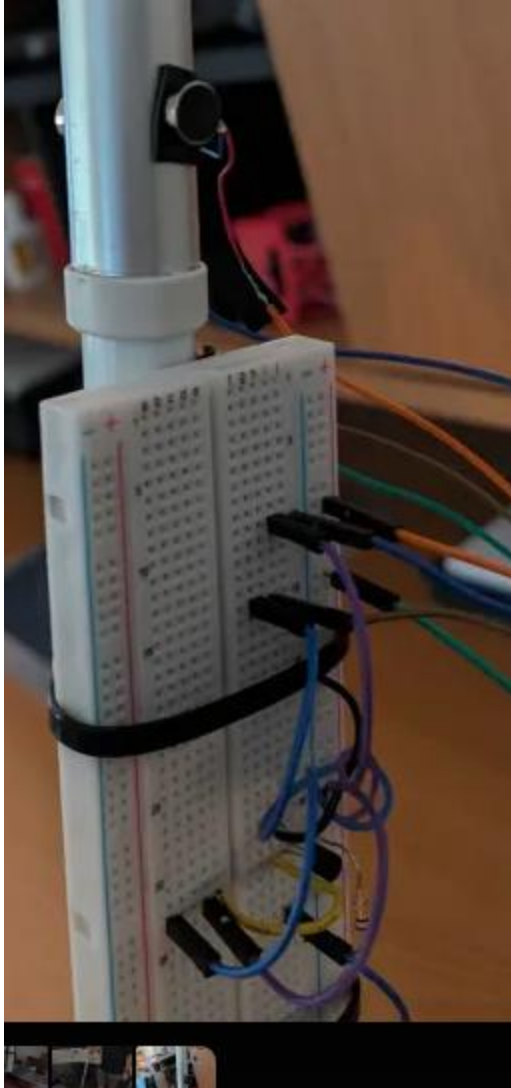


Figure 3 - Parallel breadboard circuit with resistor and transistor

DISCUSSION

Overall, we would call this project a success. We were able to complete all our outlined tasks such as the motors vibrating when the sensors detect an object as well as the motors increasing the vibration intensity as the sensors detect the object getting closer. The biggest hurdle in this project would be developing the breadboard to supply a sufficient amount of power to all the motors we planned to use without overloading the Raspberry Pi. Overloading the Raspberry Pi was a major concern of ours as when conducting further research on the motors we acquired, we realized that if we connected the motors directly to the Raspberry Pi they would overload and fry the header pins. This would have caused a big setback for us that we could not afford. Our solution to this was using a transistor and a resistor on a breadboard to mediate the current going through the Raspberry Pi. We also soldered some wires together which provided a more secure connection throughout the stick as we did not have to worry about any of the wires popping out.

FUTURE WORK

In the future, hopefully we can 3D print the stick. This would allow us to have the electronics and the power bank inside the stick. We were planning to print the stick however; the printers were unavailable. Another improvement that could be made is to have the startup time for the code be a lot shorter. When completely shut off and turned back on the walking stick takes about a 1 minute and 30 seconds to become operational. The reason for this long time startup time is due to the raspberry pi. This can be remedied by using an Arduino to make it go faster. Finally, adding sensors to the sides of the stick will let it detect objects that are on a person's left and right.